

Enterprise Business Intelligence Dashboard Specifications (Modules 7, 8, 10)

1. Executive Strategy & Dashboard Design Philosophy

In the high-stakes environment of Enterprise Resource Planning (ERP), the transition from operational data to strategic intelligence is the primary driver of competitive advantage. This document outlines the architectural specifications for the Business Intelligence (BI) layer of the `ser_ser_f6ecb7` schema. By synthesizing raw relational data from disparate modules—Branch Management, Human Resources, and Inventory—into a unified "digital twin," we provide stakeholders with a high-signal environment for decision-making.

Our "Clear & Actionable" design philosophy mandates a high signal-to-noise ratio. We eliminate visual complexity by employing simple, business-friendly naming conventions and prioritizing critical operational triggers over vanity metrics. Every visualization is engineered to serve a specific business outcome, ensuring that management moves from insight to execution with surgical precision. The following specifications establish the technical blueprint for the deployment of our core infrastructure, personnel, and asset dashboards.

2. Module 7: Branch Management Dashboard

The Branch Management Dashboard provides real-time visibility into the organization's physical footprint and operational capacity. Strategically, this dashboard monitors the geographic health of the network, ensuring that infrastructure expansion aligns with market demand and staffing efficiency. It serves as the primary dimensional filter for the entire ERP ecosystem.

2.1 Key Performance Indicators (KPIs)

The following metrics assess the health and reach of the physical network.

KPI Name	Business Meaning	Source Table (Ref)	Source Columns	Aggregation

Total Active Branches	Current operational footprint.	<code>branches</code> (Table 2.3)	<code>status</code>	<code>COUNT(id) WHERE status = 'ACTIVE'</code>
Branch Type Distribution	Logistics balance (Retail vs. Warehouse).	<code>branches</code> (Table 2.3)	<code>branch_type</code>	<code>COUNT(id) GROUP BY branch_type</code>
Regional Branch Density	Concentration by state to find underserved markets.	<code>branches</code> (Table 2.3)	<code>state</code>	<code>COUNT(id) GROUP BY state</code>
Employee-to-Branch Ratio	Evaluates staffing efficiency per location.	<code>users</code> (Table 2.68), <code>branches</code> (Table 2.3)	<code>id</code>	<code>COUNT(users.id) / COUNT(branches.id)</code>

2.2 Strategic Visualizations (Charts)

- **Branch Growth Trend (Line Chart)**
 - **What It Shows:** Monthly count of new branch activations.
 - **Why It Matters:** Visualizes expansion velocity and infrastructure scaling.
 - **Axis Definitions:** X-Axis: `DATE_TRUNC('month', created_at)`; Y-Axis: Cumulative Count.
 - **Data Sources:** `branches.created_at` (Table 2.3).
- **Geographic Distribution by State (Bar Chart)**
 - **What It Shows:** Branch count comparison across states.
 - **Why It Matters:** Highlights regional dominance and service gaps.
 - **Axis Definitions:** X-Axis: `state`; Y-Axis: `COUNT(id)`.
 - **Data Sources:** `branches.state` (Table 2.3).
- **Operational Status Breakdown (Pie Chart)**
 - **What It Shows:** Ratio of Active vs. Inactive branches.
 - **Why It Matters:** Identifies maintenance trends and network availability.
 - **Data Sources:** `branches.status` (Table 2.3).

2.3 Operational Data Tables

1. Branch Contact & Leadership Directory

- **Columns to Display:** `branch_name`, `branch_code`, `city`, `email`, `phone_number`.
 - **Source Table:** `branches` (Table 2.3).
 - **Analytical Purpose:** Direct internal communication and site-specific leadership lookup.
2. **Recent Branch Activations (Audit Table)**
- **Columns to Display:** `branch_name`, `branch_code`, `branch_type`, `created_at`, `created_by`.
 - **Source Table:** `branches` (Table 2.3).
 - **Analytical Purpose:** Accountability audit for regional expansion efforts.

2.4 Proactive Business Alerts

- **Inactive Branch Warning:** `IF branches.status = 'INACTIVE' THEN TRIGGER ALERT`. Alerts regional managers to immediate site downtime.
- **Branch Code Integrity Alert:** `IF LENGTH(branch_code) != 3 OR branch_code !~ '^[A-Z0-9]{3}$' THEN TRIGGER ALERT`. Ensures compliance with the 3-character alphanumeric standard.

The stability of the branch infrastructure provides the physical requirements for the human capital management detailed in the next section.

3. Module 8: Employee & Hiring Management Dashboard

This dashboard monitors the complete employee lifecycle, from initial requisition to active payroll liability. By tracking the talent pipeline and workforce composition, the organization can optimize production capacity and manage its single largest operational expense: personnel.

3.1 Key Performance Indicators (KPIs)

KPI Name	Business Meaning	Source Table (Ref)	Source Columns	Aggregation
Total Active Workforce	Current headcount for production capacity.	<code>users</code> (Table 2.68)	<code>is_active</code>	<code>COUNT(id) WHERE is_active = true</code>

Open Hiring Requisitions	Talent acquisition requirements.	<code>hiring_requests</code> (Table 2.9)	<code>status</code>	<code>COUNT(id) WHERE status = 'PENDING'</code>
Departmental Staff count	Functional staff distribution.	<code>users</code> (Table 2.68)	<code>department</code>	<code>COUNT(id) GROUP BY department</code>
Monthly Payroll Liability	Total estimated basic salary obligations.	<code>user_salary_details</code> (Table 2.67)	<code>basic_salary</code>	<code>SUM(basic_salary) WHERE user_id IN (SELECT id FROM users WHERE is_active = true)</code>

3.2 Strategic Visualizations (Charts)

- **Hiring Request Pipeline (Funnel Chart)**
 - **What It Shows:** Request flow from `PENDING` → `APPROVED` → `CONVERTED`.
 - **Why It Matters:** Detects recruitment bottlenecks and conversion efficiency.
 - **Data Sources:** `hiring_requests.status` (Table 2.9).
- **Employment Type Composition (Pie Chart)**
 - **What It Shows:** Breakdown of `Permanent`, `Contract`, and `Intern` staff.
 - **Why It Matters:** Evaluates labor stability and reliance on contingent staff.
 - **Data Sources:** `users.employment_type` (Table 2.68).
- **Onboarding Trend (Area Chart)**
 - **What It Shows:** New hires over time.
 - **Why It Matters:** Plans for orientation resources and tracks growth cycles.
 - **Axis Definitions:** X-Axis: `date_of_joining`; Y-Axis: `COUNT(id)`.
 - **Data Sources:** `users.date_of_joining` (Table 2.68).

3.3 Operational Data Tables

1. **Critical Hiring Requisitions**
 - **Columns to Display:** `id`, `department`, `designation`, `number_of_positions`, `expected_date_of_joining`.
 - **Source Table:** `hiring_requests` (Table 2.9).

- **Analytical Purpose:** Prioritizes HR efforts for high-volume needs (`number_of_positions > 5`).
2. **Employee Compensation Audit**
- **Columns to Display:** `u.emp_id, u.first_name, u.department, s.salary_type, s.basic_salary, s.hra, s.deductions`.
 - **Source Tables:** `users (u) JOIN user_salary_details (s) ON u.id = s.user_id`.
 - **Analytical Purpose:** Payroll verification and cross-departmental budget reconciliation.

3.4 Proactive Business Alerts

- **Delayed Joining Alert:** `IF hiring_requests.expected_date_of_joining < CURRENT_DATE AND hiring_requests.status != 'CONVERTED' THEN TRIGGER ALERT.`
- **Low Leave Balance Warning:** `IF user_leave_details.casual_leave <= 1 OR user_leave_details.sick_leave <= 1 THEN TRIGGER ALERT.` Flags potential burnout risk.

Human resources are the primary actors who manage and deploy the physical inventory and service catalogs detailed in the following module.

4. Module 10: Inventory, Products & Services Dashboard

The Inventory Dashboard ensures that the physical supply chain supports service delivery while maintaining financial and tax compliance. It aligns stock availability at the branch level with the statutory HSN and tax requirements defined in Module 9.

4.1 Key Performance Indicators (KPIs)

KPI Name	Business Meaning	Source Table (Ref)	Source Columns	Aggregation
Total Inventory Valuation	Asset value in the product catalog.	<code>inventory_products</code> (Table 2.14)	<code>total_cost</code>	<code>SUM(total_cost) WHERE status = 'ACTIVE'</code>

Low Stock Alert Count	Products below critical safety thresholds.	<code>stock_ledger</code> (Table 2.54)	<code>assets_qty, consumable_qty</code>	<code>COUNT(id) WHERE assets_qty < [System_Threshold]</code>
Active Service Catalog	Volume of active market offerings.	<code>services</code> (Table 2.44)	<code>status</code>	<code>COUNT(id) WHERE status = 'ACTIVE'</code>
Pending Stock Transfers	Logistics volume in transit between sites.	<code>stock_transfers</code> (Table 2.60)	<code>status</code>	<code>COUNT(id) WHERE status = 'IN_TRANSIT'</code>

4.2 Strategic Visualizations (Charts)

- **Stock Category Breakdown (Bar Chart)**
 - **What It Shows:** Comparison across categories (Chemicals, Machines, Tools).
 - **Why It Matters:** Visualizes capital concentration in specific asset types.
 - **Data Sources:** `inventory_products.category` (Table 2.14).
- **Service Price Distribution (Bar Chart)**
 - **What It Shows:** Count of services by `price_type` (Fixed vs. Area).
 - **Why It Matters:** Analyzes revenue model diversity.
 - **Data Sources:** `services.price_type` (Table 2.44).
- **Stock Movement Absolute Volume (Line Chart)**
 - **What It Shows:** Daily transactional activity.
 - **Why It Matters:** Identifies peak logistics periods for warehouse staffing.
 - **Data Sources:** `SUM(ABS(quantity_delta))` from `stock_movement_logs` (Table 2.55).

4.3 Operational Data Tables

1. **Product Tax Mapping Audit**
 - **Columns to Display:** `product_name, hsn_code, base_price, tax_amount, total_cost`.
 - **Source Table:** `inventory_products` (Table 2.14).

- **Analytical Purpose:** Financial validation; ensures `base_price + tax_amount = total_cost`.
2. **Branch Stock Ledger**
- **Columns to Display:** `b.branch_name, p.product_name, sl.assets_qty, sl.consumable_qty, sl.in_transit_qty`.
 - **Source Tables:** `stock_ledger (sl) JOIN branches (b) ON sl.branch_id = b.id JOIN inventory_products (p) ON sl.product_id = p.id`.
 - **Analytical Purpose:** Real-time stock visibility for service dispatchers at the site level.

4.4 Proactive Business Alerts

- **High-Value Transfer Alert:** `IF SUM(inventory_products.total_cost) > [Threshold] FOR stock_transfer_id THEN TRIGGER ALERT`. Prevents high-risk asset movement without oversight.
- **Warranty Expiry Warning:** `IF CURRENT_DATE >= (contract_start_date + INTERVAL '1 month' * services.warranty_months) - INTERVAL '30 days' THEN TRIGGER ALERT`. Triggers renewal outreach.

The integration of branch data, workforce metrics, and inventory visibility ensures that these three dashboards function as a cohesive command center for ERP efficiency.

5. Data Integrity & Documentation Standards

This document serves as the final technical blueprint for BI development within the `ser_ser_f6ecb7` schema. Adherence to the following standards is mandatory:

1. **Schema Grounding:** All dashboard components must use the exact table and column names specified in the Technical Reference (Section 2.0).
2. **Aggregation Standards:** Logic must explicitly use SQL-compliant aggregations (SUM, COUNT, DATE_TRUNC) as defined in the KPI sections.
3. **Join Constraints:** Cross-module reporting (e.g., Module 8 to Module 7) must utilize the verified foreign key relationships (e.g., `user_branches.branch_id` to `branches.id`).
4. **Signal Priority:** Developers are prohibited from adding non-specified fields that reduce the signal-to-noise ratio of the final output.